

4 EXPERIMENT 4 Human Reproductive System

Aim

To study structure and functions of human reproductive system.

Requirements

Models or photographs of the human male and female reproductive system.

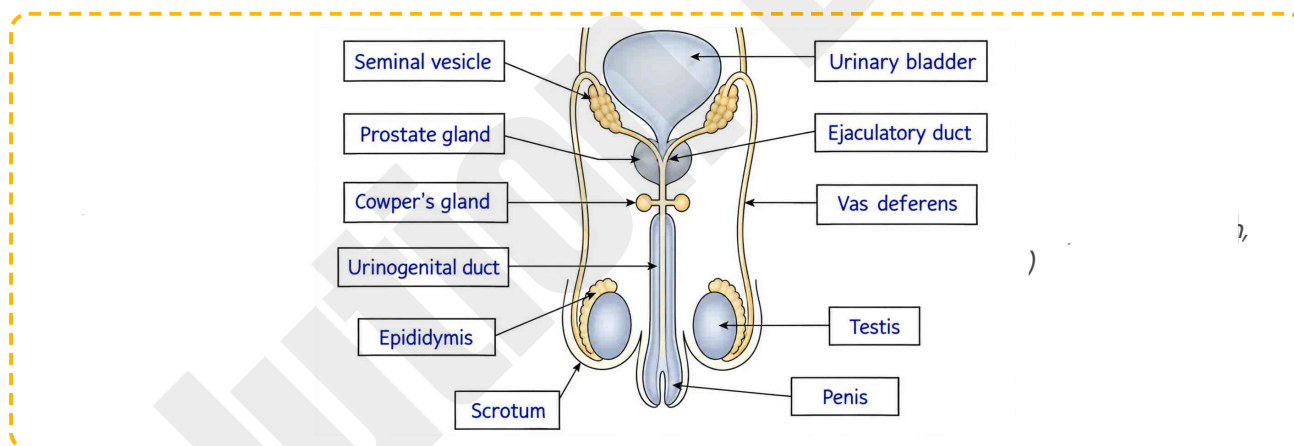
Procedure

Observe the structure of the male and female reproductive systems using the models or photographs available in the laboratory. Note down the structural and functional details of each part, and enter this information into the tabulation.

1. Human Male Reproductive System

Diagram

(Label the different parts in the given diagram)



Observation Table

(Complete the given observation table)

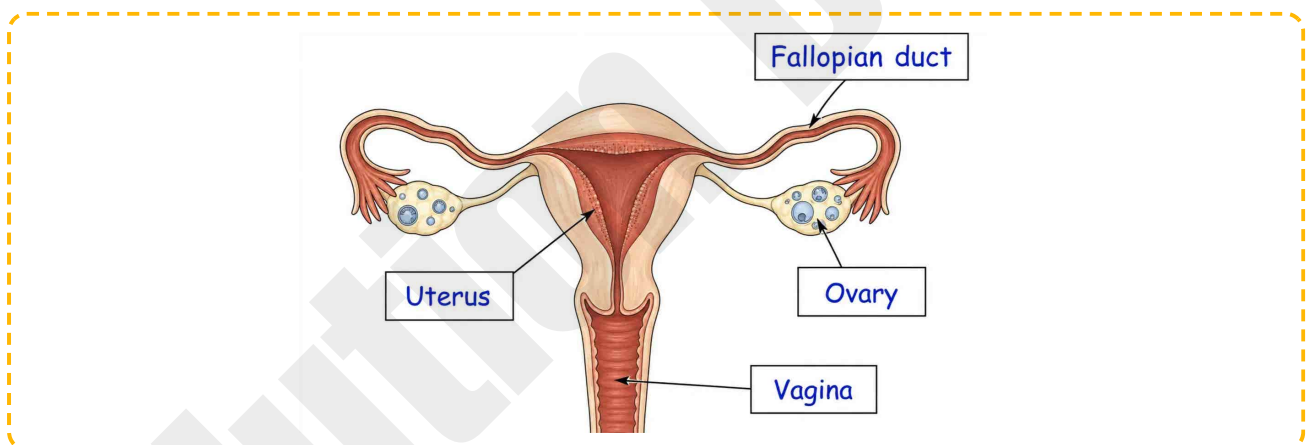
Organ	Structure	Function
Testis (Pair)	Location: Located below the abdominal cavity in the scrotum.	(1) Contain numerous seminiferous tubules. Each has germinal epithelium. The cells of germinal epithelium divide meiotically to produce sperm. (2) Testes also produce testosterone.

Different tubules / ducts	2 rete testes 2 vas efferens 2 epididymis 2 vas deferens 1 ejaculatory duct 1 urethra	(1) Sperms travel from one tubule to next. During this, they become mature and able to fertilize the ovum. (2) Which out of these ducts are cut off while performing family planning operation? In family planning operations, vas deferens are cut off so that sperms do not enter the semen.
Accessory glands associated with male reproductive system.	(1) 2 seminal vesicles (2) Single prostate gland (3) 2 Cowper's glands or Bulbourethral gland	Secretions released in urethra. All secretion + sperms = semen
Penis (Urethra passes through Penis)	Passage of urine as well as sperms.	(1) Penis transfers the semen during intercourse. (2) Urine stored in urinary bladder is released in Urethra running through penis.

2. Human Female Reproductive System

Diagram

(Label the different parts in the given diagram)



Observation Table

(Complete the given observation table)

Organ	Structure	Function
2 Ovaries	Situated in lower abdominal cavity.	Produce ova, secrete female hormones.
2 Fallopian ducts (called oviducts in lower animals)	Consists of three parts. (1) Infundibulum	Transport of ovum to uterus.

	(2) Ampulla (3) Isthmus	
1 Uterus	In the middle of lower abdomen.	(1) Development of foetus (2) Birth process
1 Vagina	The passage of uterus to outside.	(1) Passage for intercourse (2) Acts as birth canal (3) Passage for menstrual flow
Bartholin's glands	In the vaginal wall.	For lubrication and protection of vagina.

Inferences

1. The important function of human reproductive system is to produce gametes and hormones.
2. The male hormone, testosterone is produced by testis.
3. The two important female hormones estrogen and progesterone are produced by ovaries.
4. The sperm and ovum are produced by meiotic division.
5. There are 23 chromosomes each in sperm and ovum. By fertilization of these gametes zygote is formed which contains 46 chromosomes.

Multiple Choice Questions

1. Which gland in human male reproductive system is single?

- (A) Testis (B) Prostate gland
(C) Epididymis (D) Cowper's gland

Answer: (B) Prostate gland

2. Which sugar is present in the semen?

- (A) Fructose (B) Glucose
(C) Maltose (D) Lactose

Answer: (A) Fructose

3. Which sex chromosomal combination in the zygote is responsible for the formation of testes?

- (A) XX (B) XO (C) XXY (D) XY

Answer: (D) XY

4. Which part of the human female reproductive system is under the direct control of pituitary hormones?

(A) Uterus (B) Vagina (C) Ovary (D) Fallopian ducts

Answer: (C) Ovary

5. Why does women get menopause, but man does not have such stage?

(A) Female hormones stop working in 45+ age, but male hormones' secretion do not cease.

(B) Females are weak, and males are strong.

(C) Females cannot sustain the pregnancy in late age and hence they get menopause. Males have no such problem.

(D) Males have better food and exercise, female do not get such facilities.

Answer: (A) Female hormones stop working in 45+ age, but male hormones' secretion do not cease.

6. Which part of female reproductive system that provides the space for fertilization of the egg?

(A) Uterus (B) Fallopian ducts (C) Vagina (D) Ovary

Answer: (B) Fallopian ducts

7. Which group of hormones are essential for a female to become a mother?

(A) Testosterone, estrogen and progesterone

(B) Estrogen and progesterone

(C) FSH, LH, LTH, estrogen and progesterone

(D) ICSH, LH and FSH

Answer: (C) FSH, LH, LTH, estrogen and progesterone

Remember

1. Reproductive system of male continues to work even in old age.
2. Reproductive system of female works only till the menopause.

Date: _____

Teacher's Signature: _____